

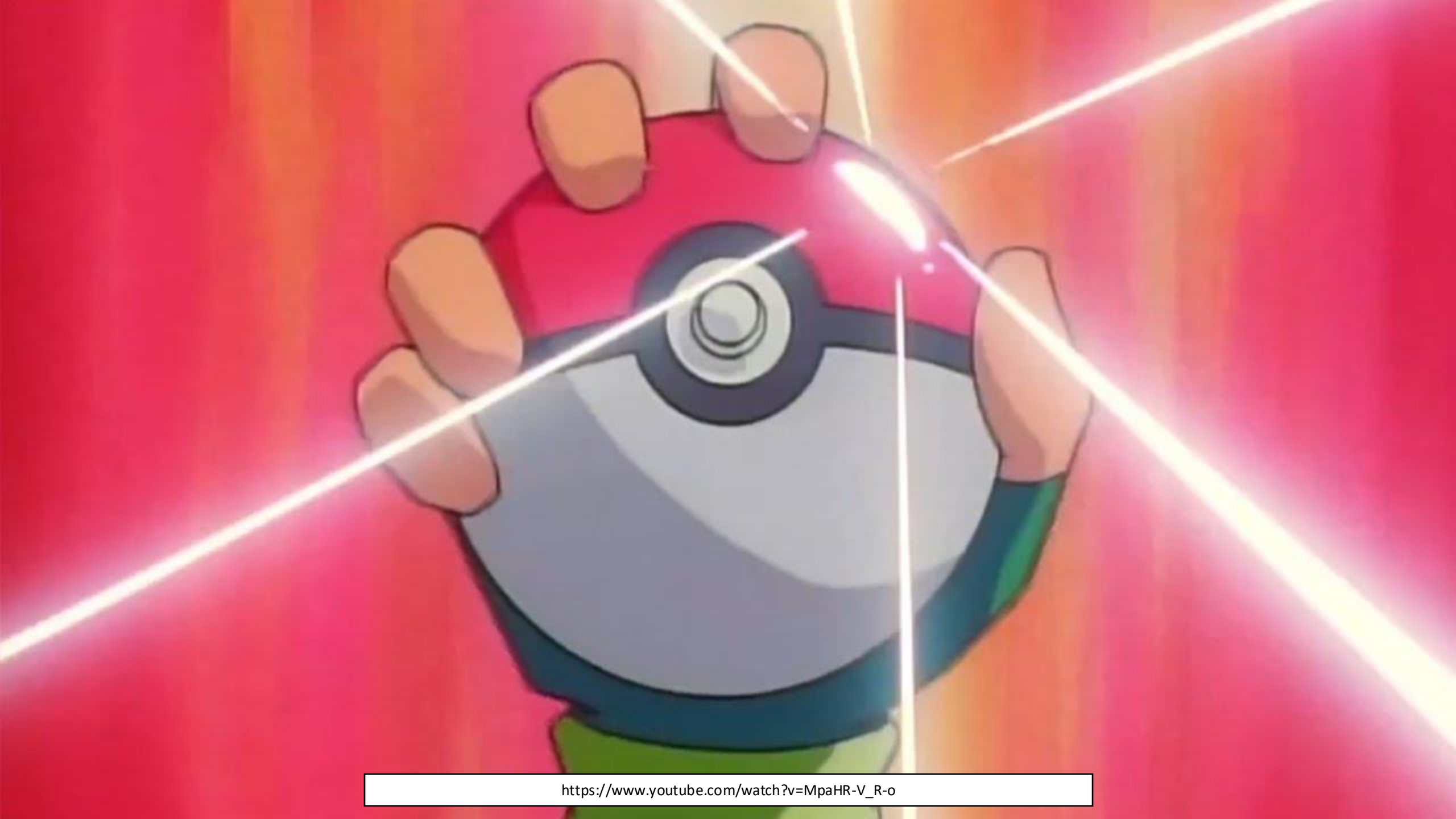


IEA-P – DEPARTAMENTO DE PROJETOS
(PROJECT DEPARTMENT)

Means of Compliance and Success Criteria

[2025]

Prof. Dr. Christopher S. Cerqueira





Fevereiro

S	T	Q	Q	S	S	D
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
1	24	25	26	27	28	

Março

S	T	Q	Q	S	S	D
1					1	2
2	3	4	5	6	7	8
3	10	11	12	13	14	15
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6	31					

Abril

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Maiο

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Junho

S	T	Q	Q	S	S	D
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Ex1	23	24	25	26	27	28
Ex2	30					

Julho

S	T	Q	Q	S	S	D
Ex2	1	2	3	4	5	6
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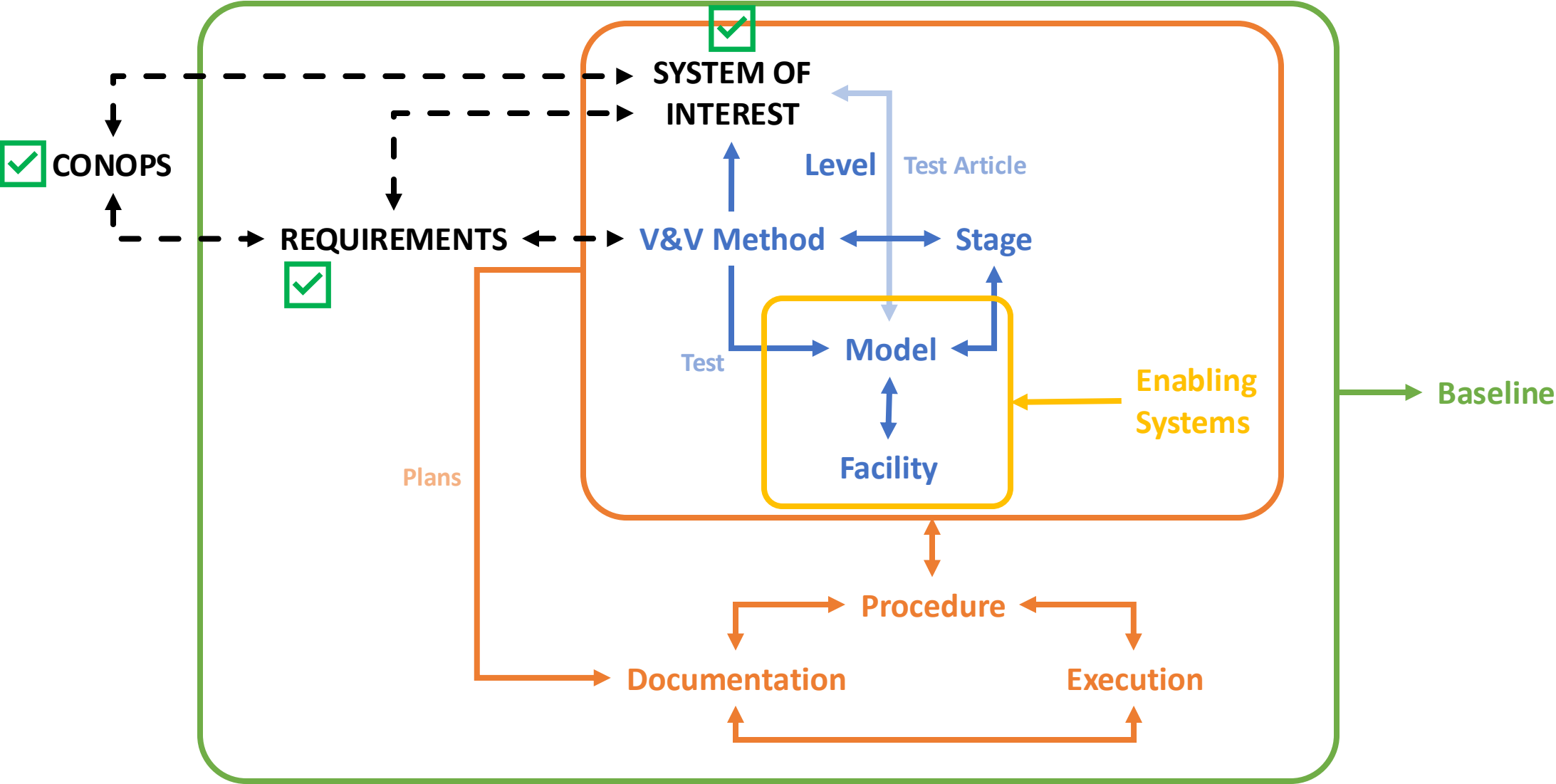
WEEK	CLASS ACTIVITY	INDIVIDUAL	W	GROUP	W
1 27-Feb 28-Feb	Course Structure and Initial Definitions Systems Engineering Review	IA-01 - Reading and Conceptual Questions (10)	5%	GA-01 - Define Groups	0%
2 06-Mar 07-Mar	Requirements Engineering Review	IA-02 - Reading and Conceptual Questions (10)	5%	GA-02 - Requirement writing and requirement validation (cross-check inter groups)	10%
3 13-Mar 14-Mar	Risk - Hazard Analysis Review	IA-03 - Reading and Conceptual Questions (10)	5%	GA-03 - Raise overall mission risks and specific to the subsystem	0%
4 20-Mar 21-Mar	AAV Introduction	IA-04 - Reading and Conceptual Questions (10)	5%	GA-04 -	20%
5 27-Mar 28-Mar	Means of Compliance Success Criteria	IA-05 - Reading and Conceptual Questions (10)	5%	GA-05 - Write the means of verification and the success criteria.	20%
6 03-Apr 04-Apr	Structuring the Life Cycle Technical Reviews	IA-06 - Reading and Conceptual Questions (10)	5%	GA-06 - Structure a lifecycle and the content of the reviews.	20%
7 10-Apr 11-Apr	Model Philosophy and Environmental Testings Questions?	IA-07 - Reading and Conceptual Questions (10)	10%	GA-07 - Structure the model philosophy	30%
8 17-Apr 18-Apr	P1 - Conceptual Questions and Case	IA-08 - Select 20 questions and a mini-case to build a coherent DVM (RTM)	60%	GA-08 -	
			100%		100%

theory





relações



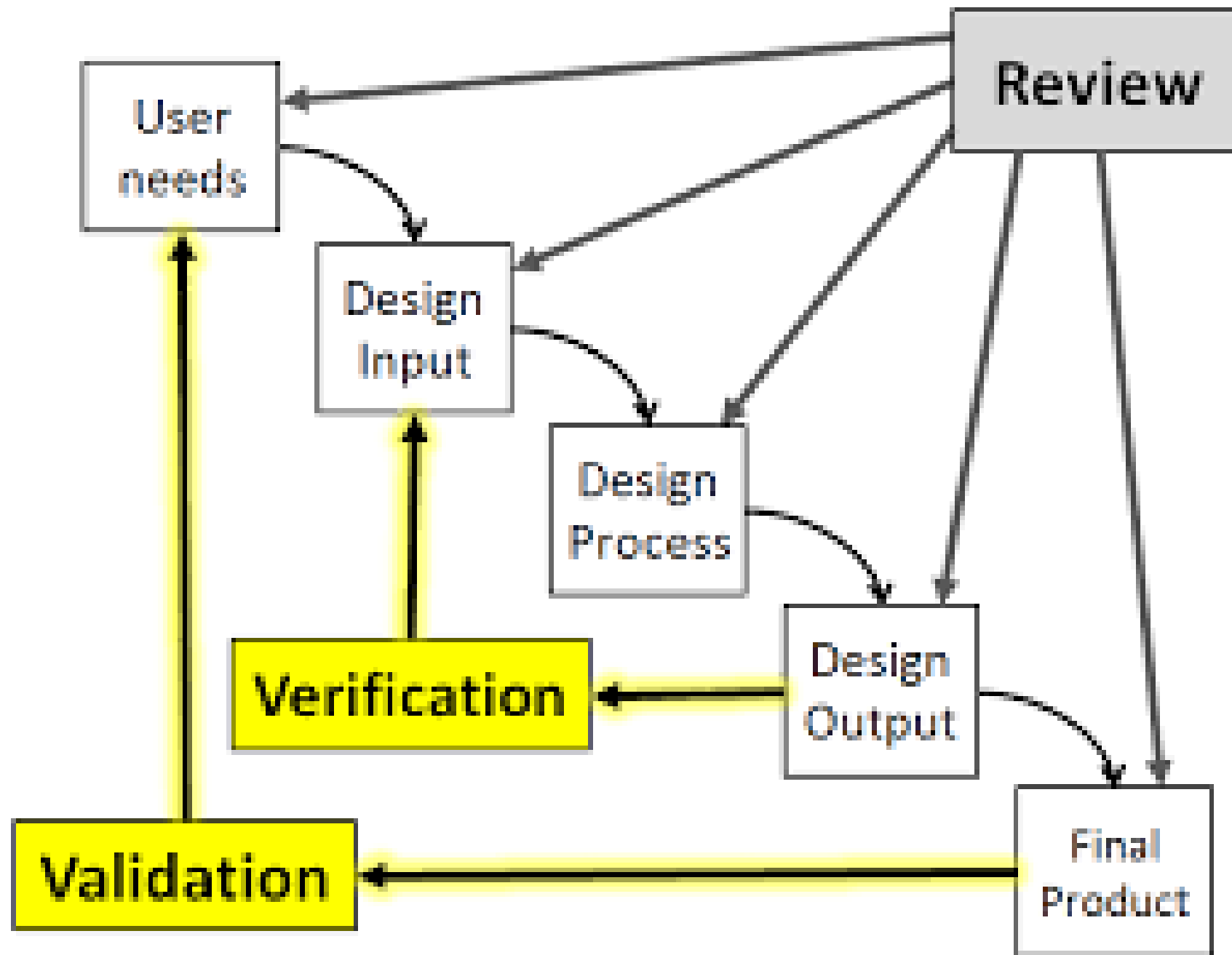


Validation x Verification

- **VERIFYING A SYSTEM:** Building the *system right*: ensuring that the system *complies with the system requirements* and conforms to its design.

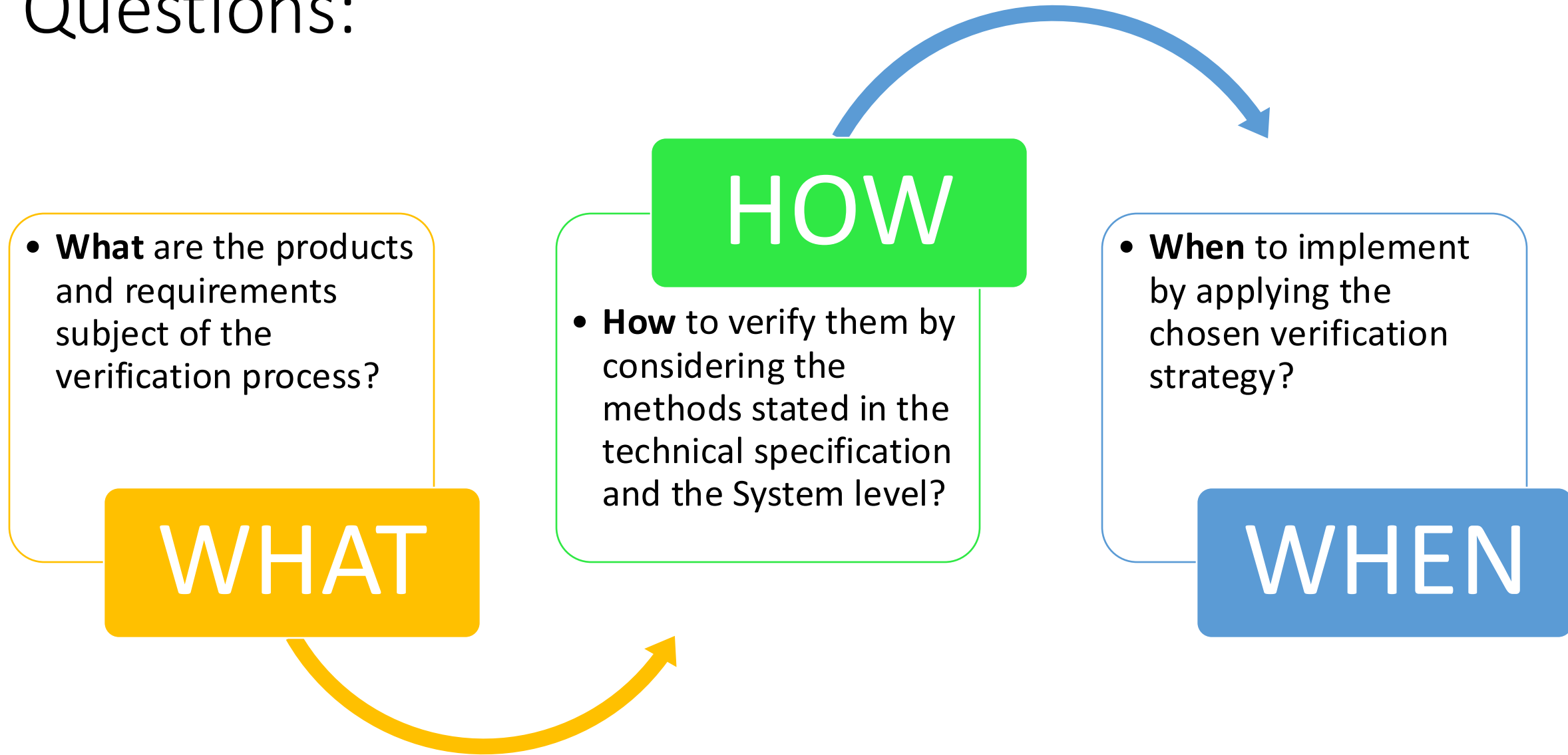


- **VALIDATING A SYSTEM:** Building the *right system*: making sure that the *system does what it is supposed to do* in its intended environment. Validation determines the correctness and completeness of the end product and ensures that the system will satisfy the actual needs of the stakeholders.





Questions:



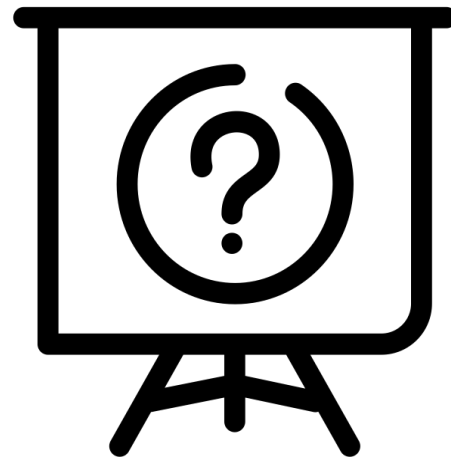


Verification & Validation Methods

“HOW”



How would you guarantee a system is working?





V&V Requirements Development

- V&V requirements should be developed concurrently with system requirements
- Systems Engineer (SE) works with Requirement Owners to help establish methodology



- On small projects, SE may try at method, phase, level, then have discipline experts' review.
- Larger projects, may use V&V Working Group or a series of sessions with Stakeholders.



SE should ensure all requirements are **verifiable as written**



Analysis

- The use of mathematical modeling and analytical techniques to **predict the suitability** of a design to stakeholder expectations based on calculated data or data derived from lower system structure end product verifications.
 - Analysis is generally used when a prototype; engineering model; or fabricated, assembled, and integrated product is not available. Analysis includes the use of modeling and simulation as analytical tools. A model is a mathematical representation of reality. A simulation is the manipulation of a model. Analysis can include verification by similarity of a heritage product.
- The use of mathematical modeling and analytical techniques to **predict the suitability** of a design to stakeholder expectations based on calculated data or data derived from lower system structure end product verifications.
 - Analysis is generally used when a prototype; engineering model; or fabricated, assembled, and integrated product is not available. Analysis includes the use of modeling and simulation as analytical tools. A model is a mathematical representation of reality. A simulation is the manipulation of a model.



Demonstration

- Showing that the use of an end product **achieves the individual specified requirement.**
 - It is generally a basic confirmation of performance capability, differentiated from testing by the lack of detailed data gathering. Demonstrations can involve the use of physical models or mock-ups; for example, a requirement that all controls shall be reachable by the pilot could be verified by having a pilot perform flight-related tasks in a cockpit mock-up or simulator. A demonstration could also be the actual operation of the end product by highly qualified personnel, such as test pilots, who perform a one-time event that demonstrates a capability to operate at extreme limits of system performance, an operation not normally expected from a representative operational pilot.
- Showing that the use of an end product achieves the **stakeholder expectations as defined in the NGOs and the ConOps.**
 - It is generally a basic confirmation of behavioral capability, differentiated from testing by the lack of detailed data gathering. Demonstrations can involve the use of physical models or mock-ups; for example, an expectation that controls are readable by the pilot in low light conditions could be validated by having a pilot perform flight-related tasks in a cockpit mock-up or simulator under those conditions.



Inspection

- The visual examination of a realized end product.
- Inspection is generally used to **verify physical design features or specific manufacturer identification**.
- For example, if there is a requirement that the safety arming pin has a red flag with the words “Remove Before Flight” stenciled on the flag in black letters, a visual inspection of the arming pin flag can be used to determine if this requirement was met. Inspection can include inspection of drawings, documents, or other records.

- The visual examination of a realized end product.
- Inspection is generally used to **validate the presence of a physical design features or specific manufacturer identification**.
- For example, if there is an expectation that the safety arming pin has a red flag with the words “Remove Before Flight” stenciled on the flag in black letters, a visual inspection of the arming pin flag can be used to determine if this expectation has been met.



Test

- The use of an end product to obtain detailed data needed to **verify performance** or provide sufficient information to verify performance through further analysis.
 - Testing can be conducted on final end products, breadboards, brassboards, or prototypes. Testing produces data at discrete points for each specified requirement under controlled conditions and is the most resource-intensive verification technique. As the saying goes, “Test as you fly, and fly as you test.”
- The use of an end product to obtain detailed data needed to determine a behavior or provide sufficient information to **determine a behavior** through further analysis.
 - Testing can be conducted on final end products, breadboards, brassboards, or prototypes. Testing produces information at discrete points for each specified expectation under controlled conditions and is the most resource-intensive validation technique.



Other types

- Review of Design
 - When verification is achieved:
 - by validation of records or
 - by evidence of validated design documents or
 - when approved design reports, technical descriptions, engineering drawings unambiguously show the requirement is met, the method shall be referred as “Review-of-design”.
- Process Control:
 - Process control values are accepted as evidence of requirements compliance.
 - Process factors are known, measured, and held to predetermined targets.
 - It is used to show dependability and consistency of process results. Process Control cannot be used to show that a system/component design complies with requirements.



Methods of V&V

4 FUNDAMENTAL METHODS	DESCRIPTION
INSPECTION	➤ Visual examination of drawings or data ➤ Examining a direct physical attribute of the item itself • dimensions • weight • physical characteristics • color or markings
ANALYSIS	➤ Evaluation of data by generally accepted analytical techniques • systems engineering analysis • Statistics • qualitative analysis • analog modeling • Similarity • computer and hardware simulation ➤ Can also be evaluation of data derived from lower-level verifications.
DEMONSTRATION	➤ Showing that the use or operation achieves the results without detailed data gathering. Utilize physical models or mockups. ➤ Often associated with the “ilities” (accessibility, transportability, serviceability, etc.)
TEST	➤ Operation of equipment during controlled conditions or when subjected to specified operational environments to evaluate performance ➤ Test is the preferred method of V&V



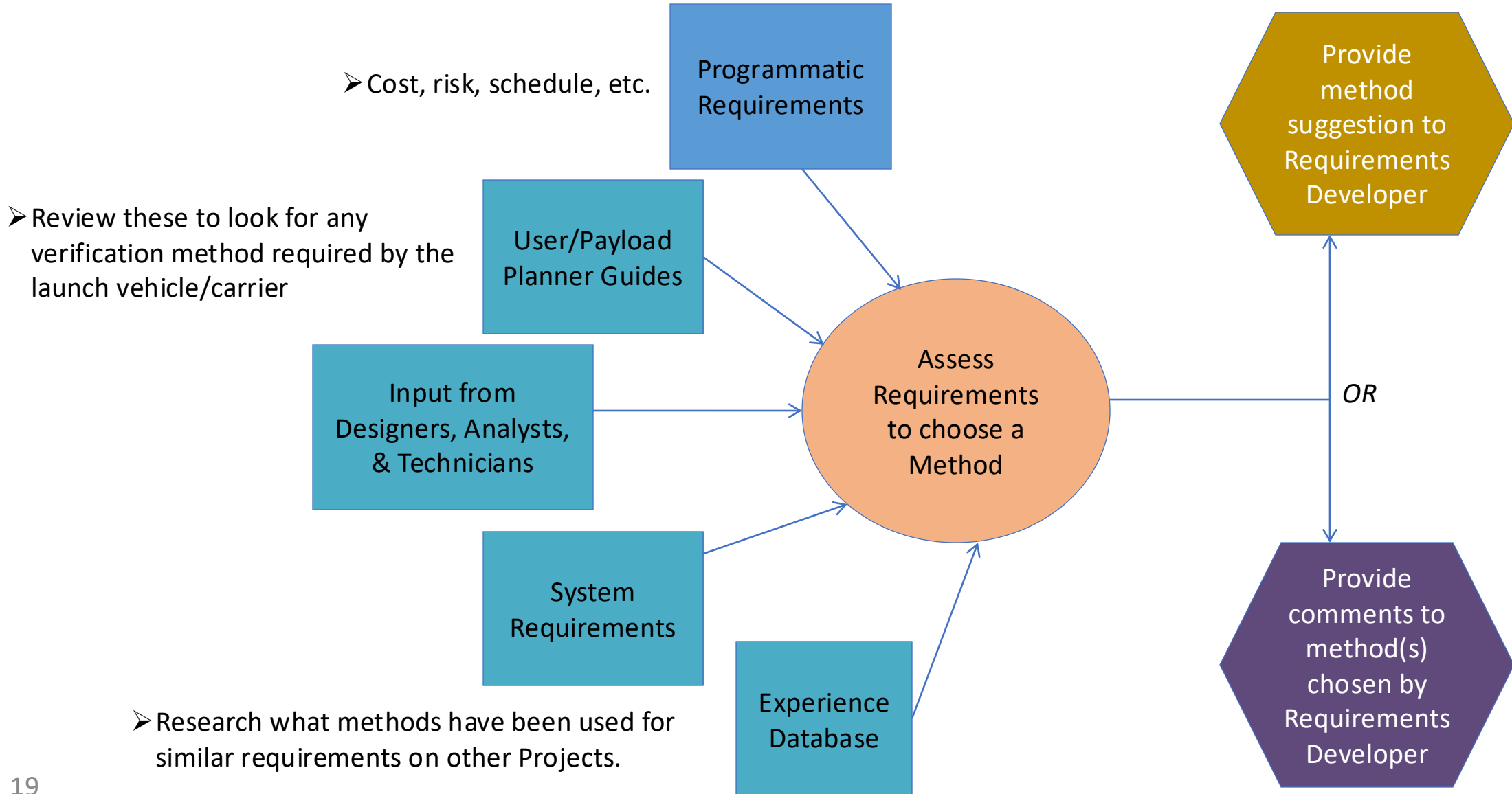
Other Methods of V&V

ADDITIONAL METHODS	DESCRIPTION
SIMILARITY	➤ Assessing by review of prior acceptance data that a system is similar or identical in design & manufacturing process to another system that has been previously qualified ➤ previous and current system predicted, or actual environments are similar ➤ Usually considered a subset of Analysis ➤ Is not used when either, or both, of the following conditions exist: • The similar item used in the assessment was itself verified using similarity as the method • Items of criticality 1 or 1R (i.e., loss of vehicle, life, or serious injury)
VALIDATION OF RECORDS	➤ The use of vendor-furnished manufacturing or processing records to ensure requirements have been met ➤ Often used for COTS products or products purchased to standards

- There are differing definitions within the community of practice.
- The key is for your Project to establish agreed-to definitions, then be consistent in using the terminology!



Choosing a V&V Method





Choosing a V&V Method

Test used when:

- *Analytical techniques do not produce adequate results
- *Failure modes exist which compromise safety, space systems or mission objectives
- *Components associated with critical system interfaces

Demonstration used when:

- *Verification of designed functions can be performed through observation
- *Results tend to be “pass/fail”, “yes/no”
- *Requirements are more subjective in nature such as human factors or maintainability
- *Uses actual or representative flight systems

Inspection is used when:

- *Drawings, documents or data can be visually checked to verify that the physical characteristics have been designed into the product
- *Typically used for design features, construction methods, workmanship, dimensions and records

Analysis is used when:

- *Accurate analytical techniques are possible
- *Test is not a cost effective option
- *Verification by inspection is not adequate
- *Analytical techniques and models have been validated



Tasks	Steps
Define verification/validation requirements.	a. Define the method by which the requirement is to be verified. (1) Test. (2) Analysis. (3) Inspection. (4) Demonstration. (5) Validation of Records. (6) Similarity. b. Define the level at which the verification/validation will occur. (1) System. (2) Subsystem. (3) Component. c. Define the phase or purpose of the verification/validation activity to be performed. (1) Development. (2) Qualification. (3) Acceptance. (4) Pre-launch. (5) Flight/Mission. (6) Post-flight. (7) Disposal.
Manage and maintain the verification/validation requirements.	a. Manage and maintain verification/validation requirements. b. Update as necessary in accordance with established data management requirement.



Success Criteria

[https://ndiastorage.blob.core.usgovcloudapi.net/ndia/2011/tes
t/11570MondayScukanec.pdf](https://ndiastorage.blob.core.usgovcloudapi.net/ndia/2011/tes
t/11570MondayScukanec.pdf)



Success Criteria

- The Product Verification/Product Validation success criteria provide **the detailed and specific criteria that determine the successful completion of the verification/validation planning activities.**
- Product Verification/Product Validation **success criteria are submitted as part of the PDR data package (ideally) and established (prepared) at least 90 days prior to the start of the verification/validation activity** to provide sufficient time to develop and publish the procedures.



1 - Define success criteria

- Develop success criteria based on the following considerations:
 - Performance criteria
 - Environment Test Limits
 - Tolerances
 - Margins
 - Specifications
 - Restrictions
 - Checkpoints
 - Effectiveness and localization



2 - Submit the completed Success Criteria for approval

- Obtain approval from the project team that the Product Verification/Product Validation Success Criteria have reached sufficient maturity to be used.
- Success Criteria should be prepared to be placed under formal configuration control in accordance with established configuration management procedures



3 - Manage and maintain the Success Criteria

- Manage and maintain the Success Criteria according to the guidance contained in the planning information.
- Success Criteria are placed under formal configuration control in accordance with established configuration management procedures.



Verification Requirements

- Inspection
- Analysis
- Modeling and Simulation
- Demonstration
- Test

Must answer
5 Questions

☒ Objective

What is the purpose of this verification?

☒ Method

What method do you need performed? What are the verification circumstances (e.g., laboratory, desk-top analysis, flight test)?

☒ Environment

What are the environmental conditions under which the item will be verified?

☒ Special Conditions (if necessary)

Are there any unique conditions (e.g., item configurations) necessary for the execution of the verification?

☒ Success Criteria

What results are to expected?



Design
Requirement

3.2.2.15.34 Recovery Force Communications

The product shall provide a communications system capable of communicating with the recovery forces pre- and post- landing

Design
Verification

Verification Objective

Pass / Fail (Success Criteria)

Perform Integrated System Test of the communications system capability to provide a voice communications and beacon with recovery forces pre and post landing within an integrated hardware / software environment

Testing will show that the communications system can transmit and receive audio at frequencies and ranges (power) represented by standard ground recovery force communications devices as defined in TBD

Perform a demonstration of the communications systems capability to provide voice and beacon communications with recovery forces pre and post landing while within a representative environment and using a production equipment configuration

Demonstration will show the ability for the communications systems to verbally communicate with the on-board communication production configuration equipment. The demonstration will also show beacon tracking within communication ranges established by TBD.

Verification Cross Reference Matrix

Paragraph #	N/A	I	A	M/S	D	T
3.2.2.15.34						VR-5T
3.2.2.15.34					VR-5D	

Traceability

SE – Translates Operational Objectives into Design Requirements

Design – Provides assessment of requirements implementation

Test – Provides assessment of requirements verifiability

SE – Provides compliance of the design requirement

Test / Implementation Group –

Ensures Verification Implementation Feasibility

Advises alternatives to support programmatics

Assesses completeness

Provides verifiability assessment

SE – Verification Allocation and Traceability Assurance



- **VR-1I: compliance of product markings shall be verified by examination of design drawings at the LRU supplier's location prior to the LRU CDR. The inspection will show that each marking on the LRU conforms to MIL-STD-130M.**
- **VR-2A: the product operational availability shall be calculated using the results of the government-accredited contractor-developed reliability and maintainability analyses performed during the design in conjunction with the design reference missions documented in report xxxx. The analysis will show that the product, in its operational environment, supported with its support equipment and personnel, across all missions, will have an operational availability of at least 97.5%.**



- **VR-3MS: Verification of the TMC's handling of 15 major and 30 minor incidents during peak hours shall be shown through a live simulation. The TMC training simulator shall be configured for peak a peak-travel-hours training class and staffed with trained TMC operators. The training-simulator operator shall inject various combinations of major and minor incidents over the peak-travel period and the TMC performance shall be recorded digitally and using digital cameras. The simulation shall be repeated using different combinations of TMC operators and sets of incident combinations. Verification shall be achieved when the TMC handles all simulated sets of incidents with all combinations of operators with no equipment or software overloads or interrupts and with no operator overloads or interrupts.**
- **VR-4D: Removal and replacement of all LRU's shall be demonstrated on the aircraft to show that each LRU can be removed and replaced without removing any other items or moving any cables.**
- **VR-5D: Perform demonstration to provide a communications system capable of communicating with the ground command team while in a representative environment and production configuration. Demonstration will show capability to communicate with recovery forces at TBD distances in the TBD terrain environment.**



- **VR-5T: Prove that the product's communications system is capable of communicating with the ground command team by performing an integrated system test within an integrated hardware/software environment. Testing will show that the product can transmit and receive to standard ground recovery forces audio at frequencies represented by communications devices defined in (TBD).**

Verification Objective

Verification Method

Environment

Note – there are no
Special Conditions

Success Criteria



PR-1	PR-2	PR-3	PR-4	PR-5
VR-1I	VR-2A	VR-3MS	VR-4D	VR-5D
				VR-5T

Specification

Design
Requirements

Verification
Requirements

**Verification Requirements
Appear in the Same
Specification as the Product
Requirements to be Verified**

Master Verification Plan

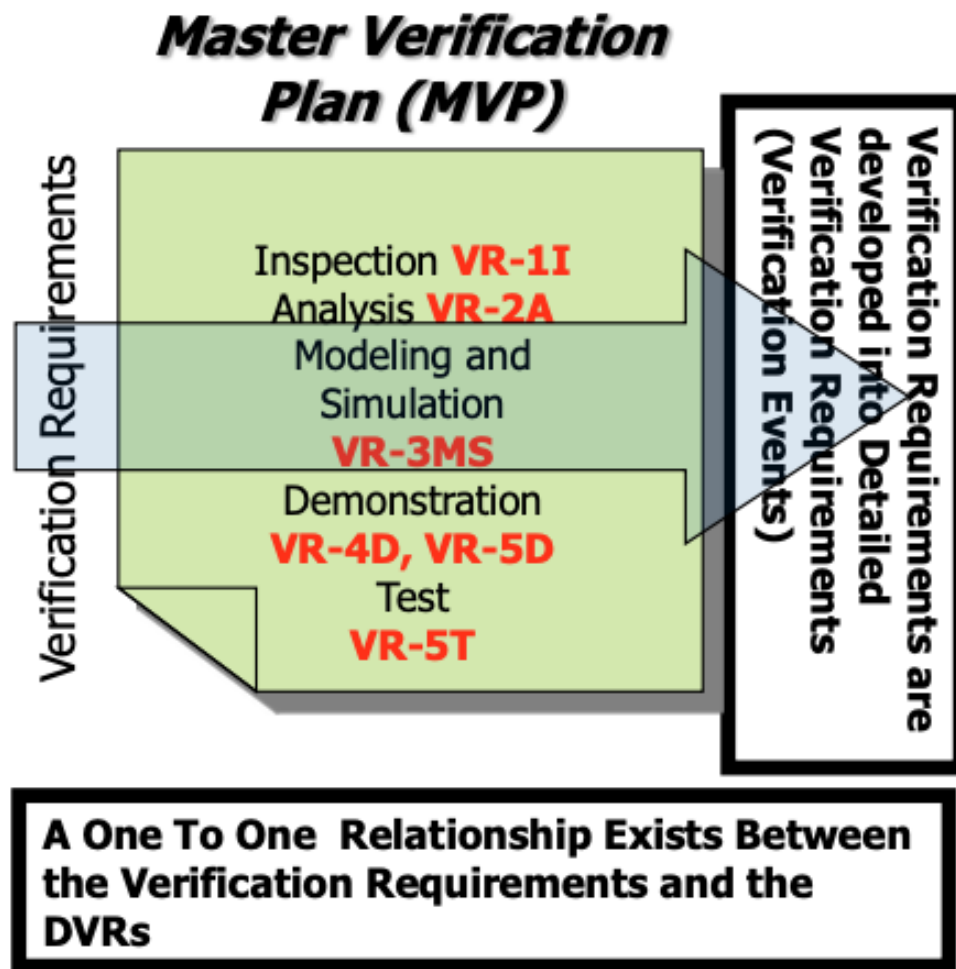
Inspection **VR-1I**
Analysis **VR-2A**
Modeling and
Simulation
VR-3MS
Demonstration
VR-4D, VR-5D
Test
VR-5T

Product Requirement	N/A	Insp	Anal	M&S	Demo	Test	Verification Requirement
PR-1		X					VR-1I
PR-2			X				VR-2A
PR-3				X			VR-3MS
PR-4					X		VR-4D
PR-5					X	X	VR-5D VR-5T

Verification Requirements



**Verification
Traceability**



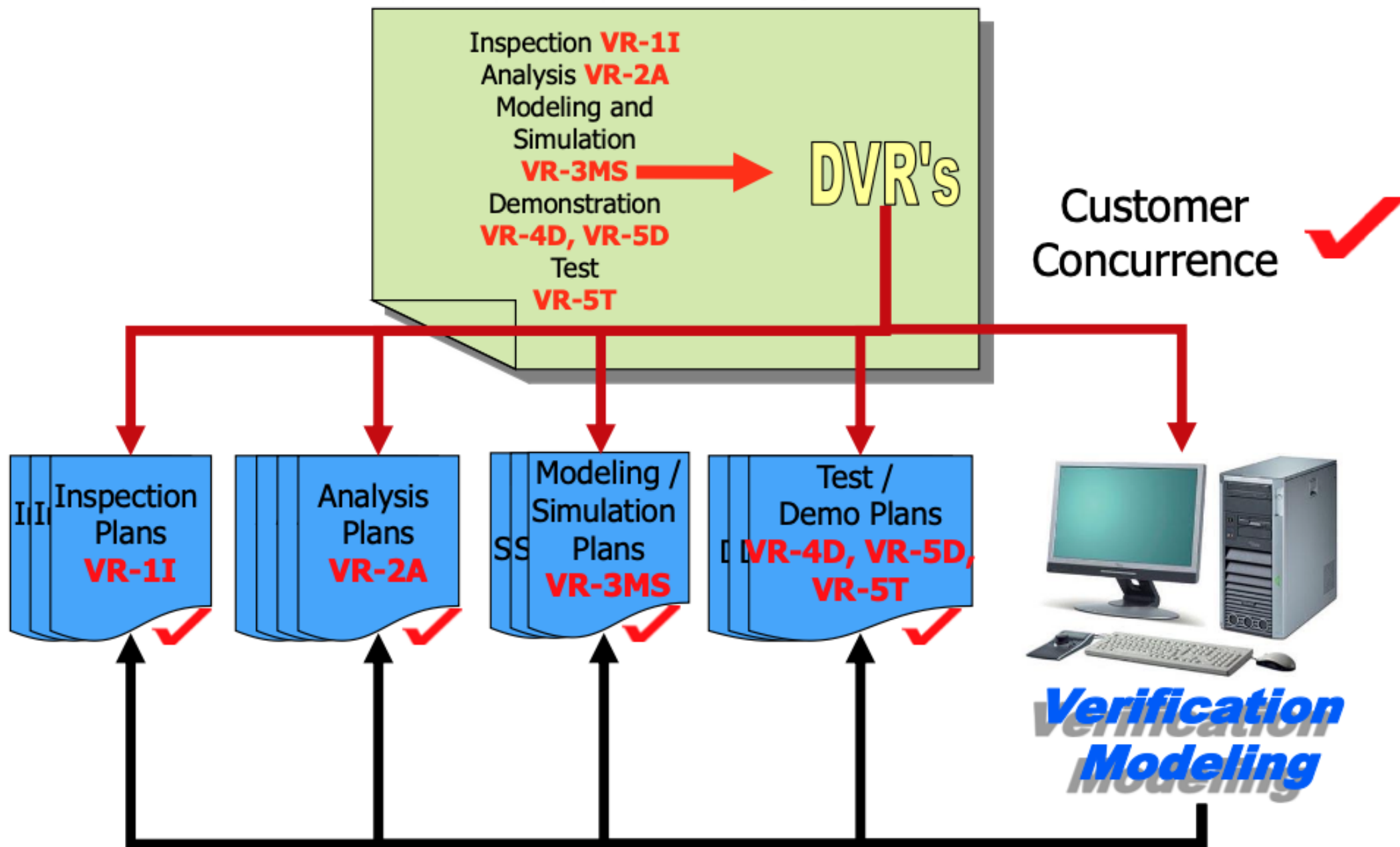
Convert verification statements into detailed verification requirements (verification events) by ----

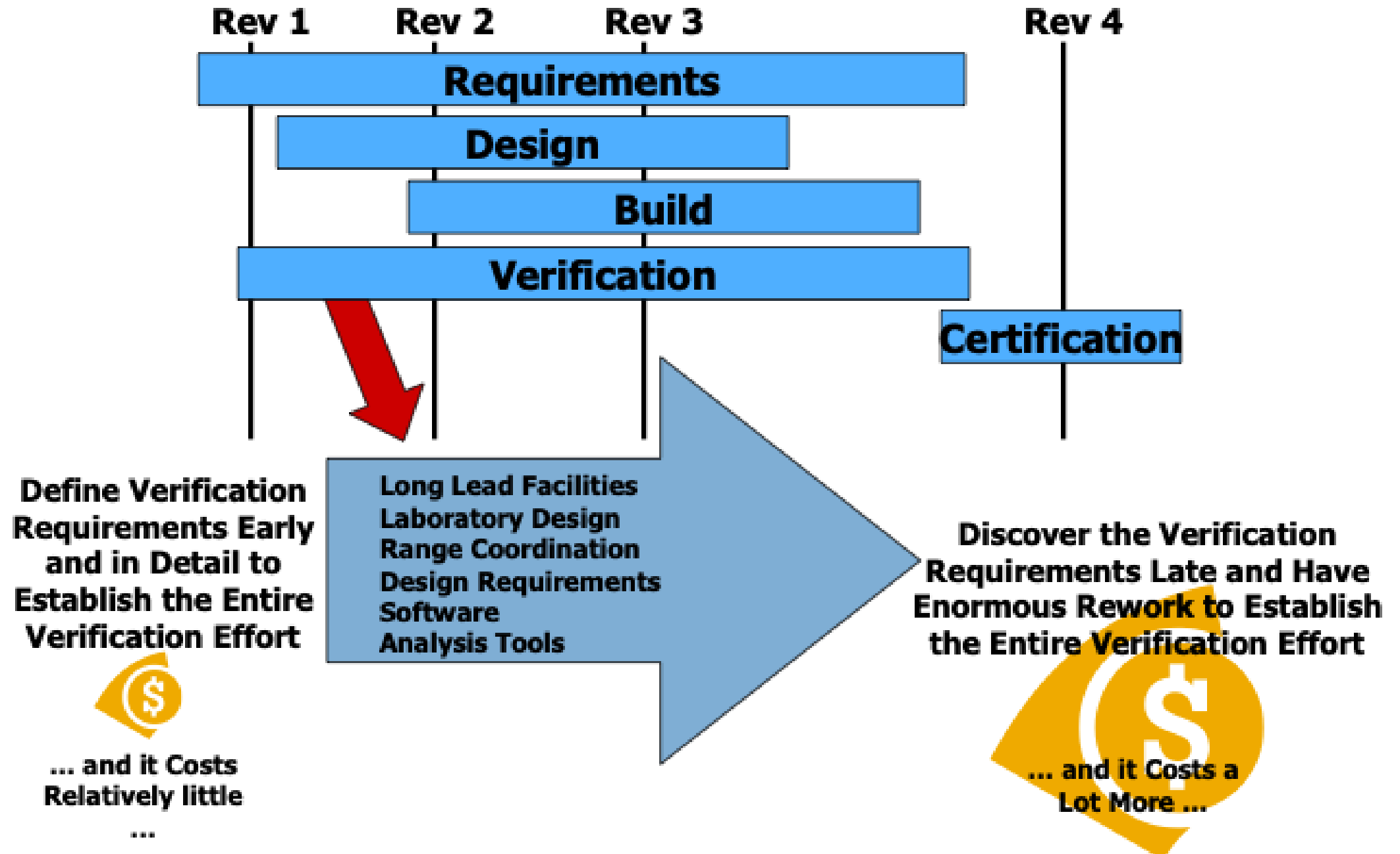
For each verification activity identified in the verification matrix, develop a detailed description of the activity including:

- **Verification configuration and its relationship to production configuration**
- **Associated prerequisites**
- **Constraints**
- **Objectives**
- **Procedures**
- **Relevant environmental conditions**
- **Pass/fail criteria- and necessary Data Set,**
- **Analysis models, if applicable.**
- **Sequence if applicable**
- **Verification Environment (e.g., Lab, Flight, Production)**



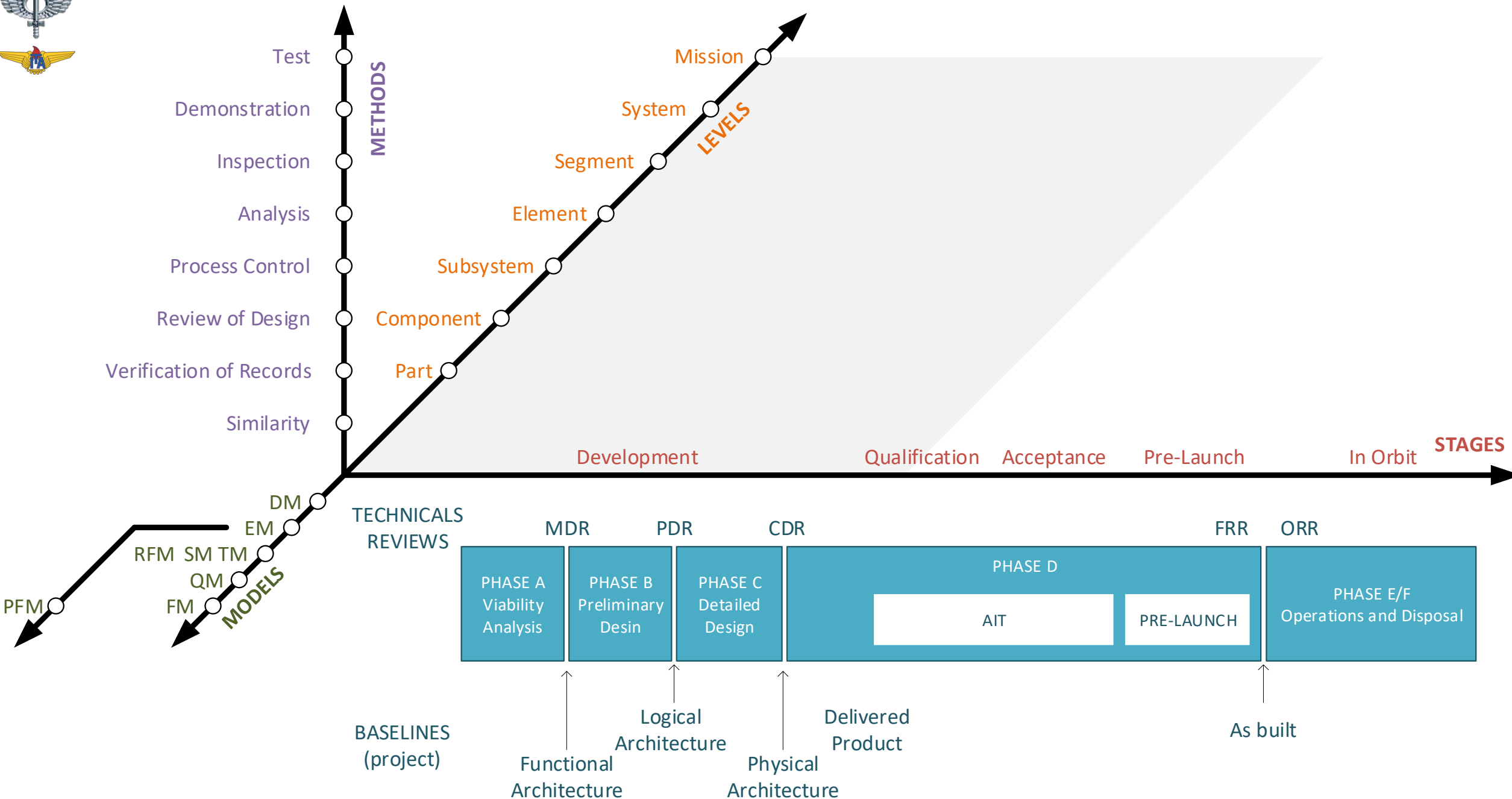
Master Verification Plan







Final Remarks

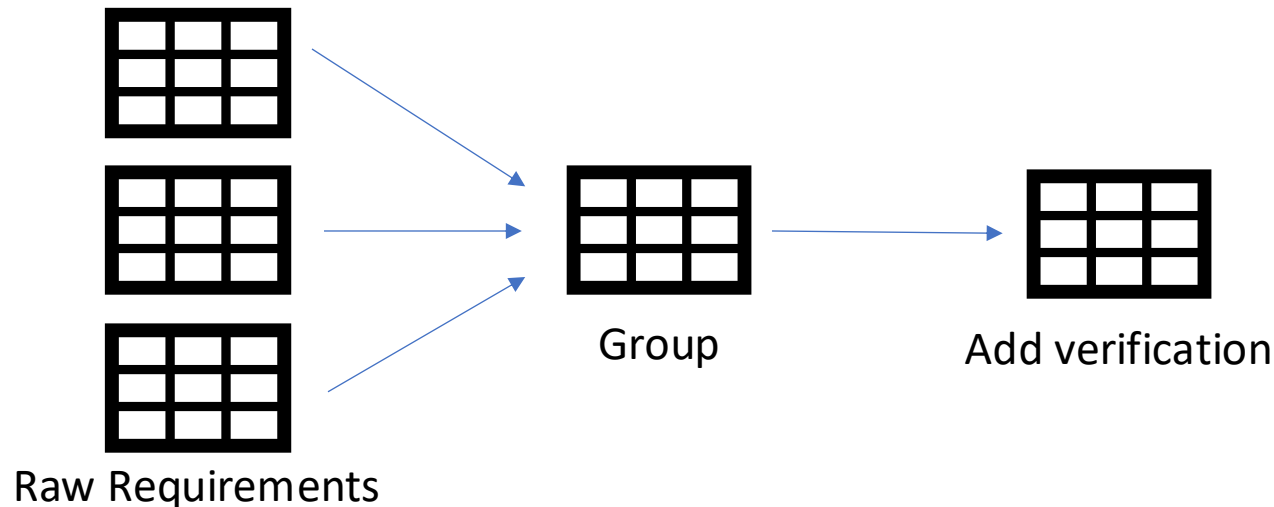






Assignments

- Get the Requirements
- Add the **Verification Method & Verification Success Criteria**
- Review / Reorganize in one spreadsheet adding the **levels**
- Indicate which **System element (mis/sys/subsys/comp)** will be involved



Grading Criteria:

1. Coherent Method x Success
2. Level Reorganization
3. Reading of the function and indicates the target subsystem in subsystem level (xx xx)
4. Suggestions for rewriting requirements, to be verifiable



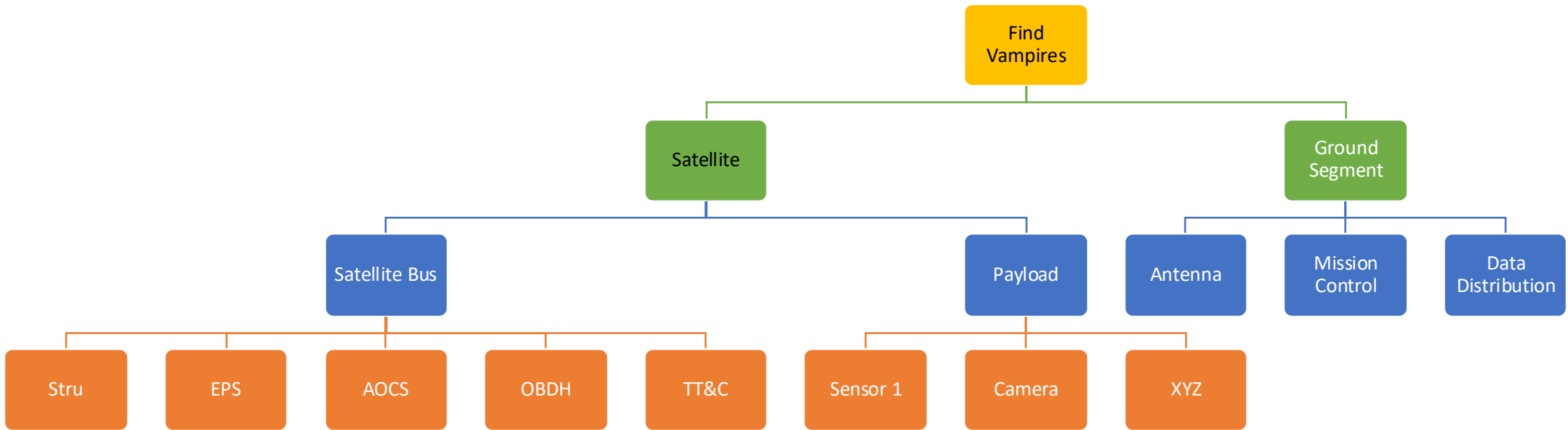
PBS

Mission →

System →

Subsystem? →

Component? →





WHAT YOU MUST DO:

1. CREATE THE CONOPS
 1. Get/Draw/Describe the concept of operations.
2. Review 15 **REQUIREMENTS WITH THEIR RATIONALES**
 1. Place 3 requirements on Mission (user) Level that will relate to your Systems, 5 on System (System of Interest) Level (2 functional at least), and 7 on Subsystem Level
3. WRITE THE V&V DATA TO EACH REQUIREMENT
 1. Indicate how each requirement will be verified,
 2. Success Criteria ([Use the template](#))
 3. Create the DVM
4. **PREPARE A PRESENTATION**